Boundary Design in Architecture and Spatial Design and Its Correspondence to the Five-Stage Model

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1. Purpose and Question of the Investigation

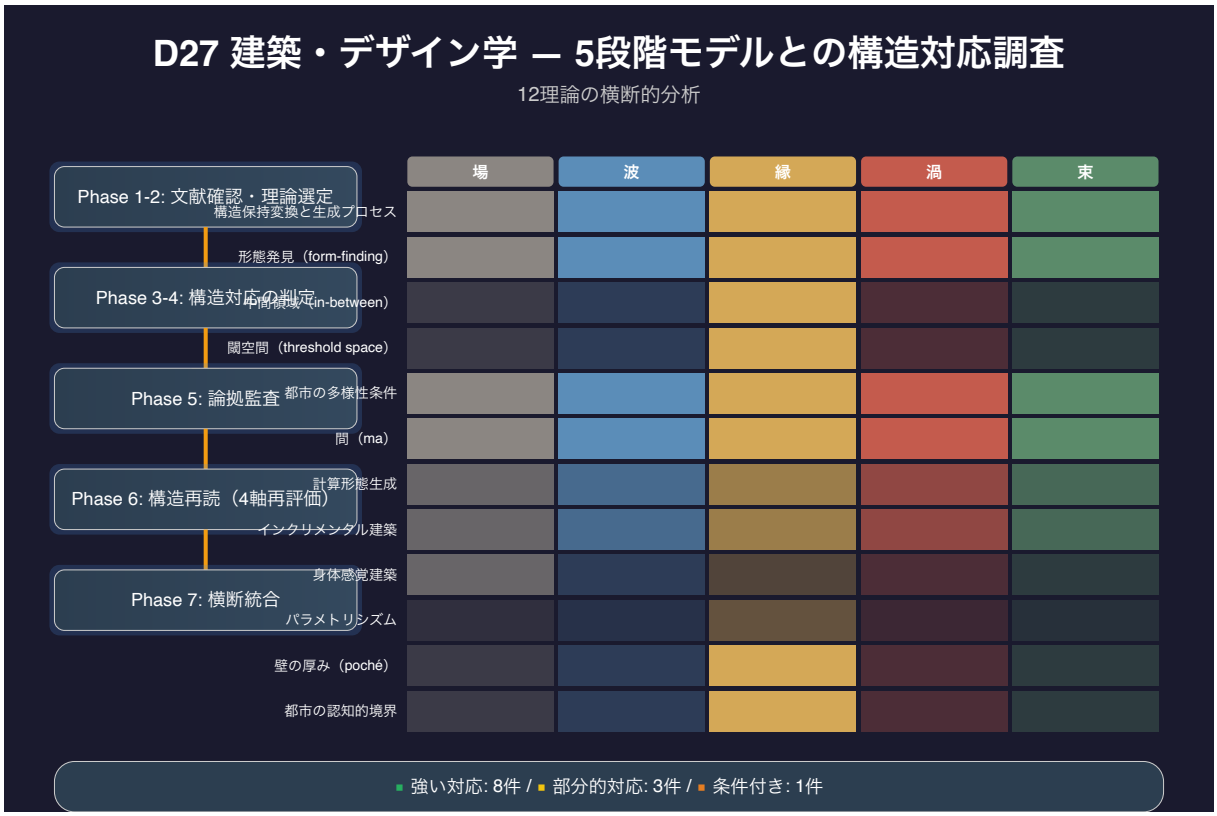

Figure 1: Boundary Design in Architecture and Spatial Design and Its Correspondence to the Five-Stage Model —

This report was prepared to answer the following question.

> Do the theories in this academic domain correspond structurally to the five-stage model (Field ( ) → Wave ( ) → Edge ( ) → Vortex ( ) → Bundle ( ))?

It systematically evaluates structural similarities between the five stages and concepts of order generation, morphogenesis, and boundary in architecture, urbanism, and spatial design. This domain aligns particularly closely with the concept of Edge because architectural theory has consistently moved in the direction of "designing boundaries as thickness, gradient, and activity." Three theoretical currents intersect here: (1) dynamic morphogenesis (Otto), (2) structure-preserving transformation (Alexander), and (3) the design of boundaries (van Eyck / Hertzberger). In addition, the report assesses twelve theories, including six candidates for promotion to CA status: urban diversity (Jacobs), Japanese space-time ( ), computational morphogenesis (Menges), incremental architecture (Aravena), embodied architecture (Pallasmaa), and parametricism (Schumacher), as well as wall thickness (poché) and cognitive urban boundaries (Lynch).

This report presents the findings of an investigation. It is not a document written to justify any particular theory. Areas of strong correspondence and areas requiring reservation are described with the same level of detachment.

2. Method of Investigation

Scope and Selection Criteria

In this domain, twelve theories and practical models were selected for evaluation. They were chosen according to three criteria: (1) they explicitly address processes of generation or transformation; (2) structural analysis is possible on the basis of primary literature or well-established secondary literature; and (3) they enable comparison across different lineages within the domain.

Evaluation Method

For each theory, the report first explains what the theory describes, and then examines where correspondences with the five stages can be found. Judgments of correspondence are based on a comprehensive assessment of textual concepts, concrete cases, connections between stages, whether there is a circulation from Bundle into the next Field, and the risk of overinterpretation.

Criteria of Judgment

- **Strong structural correspondence**: the core concepts of the theory correspond concretely to multiple stages of the five-stage model, and the main text can explain why this is so
- **Conditional correspondence**: correspondence can be indicated, but interpretive reservations remain in the mapping onto the stages
- **No confirmed correspondence**: there are partial approximations, but the theory cannot be read naturally as the structure of the five stages as a whole

Limits of the Investigation

This investigation does not examine the complete body of work of each theory exhaustively. It focuses on the core concepts of major works and extracts structure at a level of granularity suitable for comparison. As a result, some areas do not fully address internal theoretical details or historiographical disagreements.

3. Overview of the Model

The five-stage model describes the creative process in the following five stages.

Stage	Definition
Field	An undifferentiated state. It refers to initial conditions in which neither direction nor structure has yet been fixed.
Wave	A stage of exploration in which multiple directions diverge and differences are amplified.
Edge	A stage of tension in which opposing elements coexist without immediately converging into either side.
Vortex	A stage in which a new coherence or order rises out of tension.
Bundle	A stage in which form becomes fixed and stabilizes as a reusable structure.

This model is not a one-way line. It includes a circulation in which Bundle becomes the condition for the next Field. The five stages are also not strictly separated boxes, but are understood

as a continuous spectrum. The model includes the hypothesis that the same structure can be repeated across different scales, from momentary thought to institutional formation to long-term historical change.

If a sense of discomfort remains at this point, that itself is an important perspective for this investigation. Readers may continue while retaining that discomfort.

4. Results: Overall Picture

Figure 2: Boundary Design in Architecture and Spatial Design and Its Correspondence to the Five-Stage Model — ×5

List of Evaluated Theories

Theory	Proponent	Structural Correspondence
Pattern Language / Nature of Order: mutual support among centers and unfolding (Alexander)	Alexander, C. (2002-2005) *The Nature of Order* 4 , Center for Environmental Structure; Alexander, C. et al. (1977) *A Pattern Language*, Oxford University Press	Strong

Theory	Proponent	Structural Correspondence
Finding Form: exploration of equilibrium forms through physical models (Frei Otto)	Frei Otto (1925-2015), Institute for Lightweight Structures (IL), Stuttgart	Strong
in-between realm / doorstep: boundary as mediation between binary oppositions (van Eyck)	Aldo van Eyck (1918-1999), Team 10 CIAM Otterloo 1959	Strong
competence / threshold space: designing gradients between public and private (Hertzberger)	Hertzberger, H. (1991) *Lessons for Students in Architecture*, 010 Publishers	Strong
Conditions of urban diversity: boundary permeability and the ecology of streets (Jacobs)	Jacobs, J. (1961) *The Death and Life of Great American Cities*. Random House. : (2010,)	Strong
Spatial theory of (ma): an architectural account of Japanese space-time	(1978,); (1999)	Strong
Computational morphogenesis: form generation through materials and computation (Menges)	Menges, A. Institute for Computational Design and Construction (ICD), Stuttgart ; Oxman, N. MIT Media Lab Mediated Matter Group	Strong
Half-house / incremental architecture: progressive generation over time (Aravena / ELEMENTAL)	Aravena, A. / ELEMENTAL. Quinta Monroy (2004, Iquique, Chile)	Strong
Bodily sensation and architecture: phenomenology of tactile perception and spatial experience (Pallasmaa)	Pallasmaa, J. (1996/2005) *The Eyes of the Skin: Architecture and the Senses*. Wiley.	Conditional
Parametricism: continuous differentiation and the logic of fields (Schumacher)	Schumacher, P. (2008) "Parametricism as Style — Parametricist Manifesto" (presented at Venice Biennale 2008); Schumacher, P. (2011-2012) *The Autopoiesis of Architecture*, 2 vols., Wiley.	Not confirmed

Theory	Proponent	Structural Correspondence
poché: wall thickness as the material band of boundary	A representational technique and concept in architecture since the Beaux-Arts tradition. Urban-theoretical development: Rowe, C. & Koetter, F. (1978) *Collage City*, MIT Press	Strong
edge: urban image boundaries and seams (Lynch)	Lynch, K. (1960) *The Image of the City*, MIT Press	Strong

Of the twelve items evaluated, ten showed strong structural correspondence, one showed conditional correspondence, and one showed no confirmed correspondence. Overall, this domain shows correspondence with the five-stage model at relatively high density, but the manner of correspondence is not uniform. Some theories provide a thick description of the conditions of Field, while others describe with precision the tensions of Edge or the fixation of Bundle.

What matters most is not the high rate of correspondence in itself, but how each theory describes particular stages through its own vocabulary. Even when dealing with the same phenomenon of "creation," this domain illuminates different aspects of the five stages through distinct layers such as the body, institutions, time, perception, practice, and symbol.

5. Results: Examination of Major Theories

5.1 Pattern Language / Nature of Order: mutual support among centers and unfolding (Alexander)

The theory "Pattern Language / Nature of Order: mutual support among centers and unfolding (Alexander)," as developed in Alexander, C. (2002-2005) *The Nature of Order* 4 , Center for Environmental Structure; Alexander, C. et al. (1977) *A Pattern Language*, Oxford University Press, holds that "life" can be perceived and compared as the degree of order shared by architecture, the city, and nature, and that its skeleton can be explained through the arrangement of centers and their interrelation, that is, wholeness (Book 1, 2002). Centers are not geometric center points, but regions of intensity established within a field-like condition, and by supporting one another, that is, by intensifying one another, they increase the coherence of the whole.

Alexander's account of wholeness as the mutual support of centers is an architectural description of Field in the five-stage model, understood as field-like centrality. The relationality through which centers intensify one another corresponds to Wave, the emergence of difference. The fact that Boundaries appear among the fifteen properties and are defined as "thickened outlines that strengthen centers" is an architectural confirmation that Edge is not merely a boundary line but a generative band. The structure-preserving transformation of unfolding corresponds directly to the operations by which the whole is strengthened without being broken at each step from Field to Wave to Edge to Vortex to Bundle. In particular, the stepwise generative process described in Book 2, beginning from an existing whole, adding transformations one move at a time, and evaluating them through deep feeling, is the dynamic process of the five stages themselves.

This belongs to architectural theory and design methodology. Alexander (1977, 2002-2005). The shift from *A Pattern Language* to *The Nature of Order* was motivated by criticism of an overly modular understanding of patterns. Reception in architecture remains polarized: some have reevaluated the work as a candidate science of cities, while others criticize the testability of the

late cosmology of Book 4. In Japanese-language contexts, *A Pattern Language* has been widely received, whereas understanding of *The Nature of Order* remains limited.

5.2 Finding Form: exploration of equilibrium forms through physical models (Frei Otto)

The theory "Finding Form: exploration of equilibrium forms through physical models (Frei Otto)," associated with Frei Otto (1925-2015) and the Institute for Lightweight Structures (IL), Stuttgart , argues that in lightweight structures, good form is discovered not through arbitrariness on the drawing board but through form-finding, that is, through the exploration of equilibrium forms generated by materials, mechanics, and boundary conditions (Otto & Rasch, 1996). Physical models such as soap films (minimal surfaces), hanging chains (catenaries), and fabric or cable nets function not as models of a finished object but as iterative tools of analysis and design.

Form-finding is a physical self-organizing version of the five-stage model. Field (an undetermined physical system) → Wave (tension and fluctuation) → Edge (boundary conditions determining form) → Vortex (the rise of equilibrium shape) → Bundle (the institutionalization of knowledge through iterative design and IL Mitteilungen). It belongs to the same family of physical self-organization as the BZ reaction and Turing patterns in D03 chemistry, and its insight that "boundary conditions determine form" resonates directly with D29 complex systems CX-001 on the boundary conditions of dissipative structures.

What is distinctive in Otto is that he made generation without consciousness or intention visible within the context of architecture. Together with D03 (chemistry), this provides powerful evidence that the five stages operate even at levels prior to consciousness. This belongs to structural engineering in architecture. Frei Otto (1925-2015), IL (founded in 1964). He received the Pritzker Prize in 2015, awarded posthumously. His research method centered on physical models continues to be cited academically, and its inheritance within computational design is ongoing. The Munich Olympic Stadium (1972) is the representative work.

5.3 in-between realm / doorstep: boundary as mediation between binary oppositions (van Eyck)

The theory "in-between realm / doorstep: boundary as mediation between binary oppositions (van Eyck)," associated with Aldo van Eyck (1918-1999) in the context of Team 10 (from the late phase of CIAM to Otterloo 1959), criticizes the binary oppositions of modern architecture, such as inside/outside, public/private, and individual/collective, and argues that it is precisely the in-between that mediates them, restores relationality, and produces humanized space. He proposed the doorstep as a planning unit. Oppositions are reconciled not by simply joining terms together, but by designing the interval between them.

The in-between is the most direct theoretical formulation of Edge in architecture. The proposition that "boundary is not a line but a region" resonates directly with D25 anthropology AN-001, according to which transitional phases are bands. "Designing the between of oppositional terms" overlaps with the operational definition of Edge in M2. The "thickness and activity" of the doorstep is architectural demonstration that Edge has spatial, temporal, and social extension.

What is distinctive in van Eyck is the position of the in-between as a design operation. This is not analysis but design. In other words, it is not only a theory that describes M2, but a bridge toward a methodology that implements M2. This belongs to architectural theory. Its lineage runs through Team 10 and the break from CIAM. As a critique of the functional segregation of modernism, it argued for the recovery of relationality and human qualities. It directly influenced

Hertzberger. In recent years, reevaluation has progressed with the publication of the collected *Writings* (2008).

5.4 competence / threshold space: designing gradients between public and private (Hertzberger)

The theory "competence / threshold space: designing gradients between public and private (Hertzberger)," presented in Hertzberger, H. (1991) *Lessons for Students in Architecture*, 010 Publishers, argues that the boundary between public and private should be designed as a staged gradient, and that the transitional zone becomes a mediator between publicness and privacy (Hertzberger, 1991). Accessibility, responsibility, and gaze are treated as design variables, and semi-public and semi-private zones are layered.

Threshold space is the inhabitation of Edge. The gradient design of public and private is an architectural implementation of the idea that Edge is a band possessing gradient. It also connects with Gradients, one of Alexander's fifteen properties. The concept of competence, according to which space gives users possibilities for action, is an architectural-pedagogical description of Edge as a dynamic field generating action rather than a static boundary.

The concept of competence is Hertzberger's distinctive contribution. The formulation that space "gives possibilities for action" is an architectural version of Gibson's affordance, and it describes in architectural vocabulary that Edge is not a passive boundary but an active generative device. This belongs to architectural education. Hertzberger (1991). He is the intellectual successor of van Eyck and part of the Dutch structuralist lineage in architecture. *Lessons for Students in Architecture* is widely read as a classic text in architectural education. Centraal Beheer (1972) is the representative work.

5.5 Conditions of urban diversity: boundary permeability and the ecology of streets (Jacobs)

The theory "Conditions of urban diversity: boundary permeability and the ecology of streets (Jacobs)," presented in Jacobs, J. (1961) *The Death and Life of Great American Cities*. Random House. : (2010,), argues that urban vitality and diversity require the simultaneous fulfillment of four conditions: (1) mixed primary uses, (2) short blocks, (3) buildings of different ages mixed together, and (4) sufficient density (Jacobs, 1961, Part Two). Boundaries such as breaks in land use, giant blocks, and highways can sever the "flow" of the city and its ecology of diversity. Jacobs described this destructive effect of boundaries as border vacuum.

Jacobs's four conditions can be read as ecological conditions under which Edge can function. Short blocks increase the density of Edge, whereas long blocks reduce it. Mixed uses increase contact among heterogeneous elements and thereby raise the frequency with which Edge emerges. Buildings of different ages create temporal multilayeredness, which is a precondition for Bundle. Sufficient density increases the probability of interaction and secures the amplitude of Wave. Border vacuum is a description of what happens when Edge dies, and thus illuminates the validity of the concept of Edge from the negative side. This also corresponds to findings in D29 complex systems, where deviation from critical states amounts to death. Sidewalk ballet is one of the most poetic and concrete urban-scale descriptions of the five stages: diverse actors spontaneously generate order on the street through a process that can be read as Field (street space) → Wave (diversity of actors and changes across time of day) → Edge (encounter and contact) → Vortex (the self-organization of safety and vitality) → Bundle (neighborhood identity).

What is distinctive in Jacobs is the description of border vacuum, that is, the empirical observation of what the absence of Edge produces. While many theories speak of the presence of Edge, Jacobs described the death of the city caused by the absence of Edge, as in highways, giant

developments, and single-use zones. In this way, she paradoxically demonstrated the necessity of the concept of Edge. This belongs to urban planning. Jacobs (1961). It is a critique of modern urban planning, especially the large-scale development represented by Robert Moses, and it had decisive influence on urban theory in the second half of the twentieth century. The concept of "organized complexity" is borrowed from Weaver (1948) and creates a point of contact with complex-systems science.

5.6 Spatial Theory of (ma): an architectural account of Japanese space-time

The theory "Spatial theory of (ma): an architectural account of Japanese space-time," associated with (1978,) and (1999) , understands as a framework for grasping space and time as an inseparable spatiotemporal experience. Whereas the "space" of Western architecture abstracts space as three-dimensional geometry, includes time, the body, and action (, 1978 exhibition catalogue). In architecture, is concretized as the interval between columns, as variable spatial partitioning through fittings, as continuity and discontinuity between inside and outside in spaces such as engawa and doma, and as visual and bodily relations with the garden.

shows the most comprehensive correspondence as a Japanese cultural expression of the five stages. The affinity between Stage 1, Field understood as emptiness or drift, and the open interval quality of , in which the absence of something is itself something, is a unique connection not found in other architectural theories. Whereas Western architectural theory primarily describes Stage 3, Edge as boundary, has the scope to cover the entire five-stage sequence, from Stage 1, Field, through Stage 5, Bundle, as a concept running through all of them. The engawa is one of the most concrete architectural implementations of Edge. The fact that the permeability of the boundary can be dynamically controlled by opening and closing fittings is material evidence for the variability of Edge, which is not a fixed line but a band whose quality changes with the situation.

The risk of reducing "Japaneseness" to a sign must be stated frankly. The 1978 exhibition itself packaged Japanese culture, and there is a danger that the concept of may be consumed as an exoticism of Japanese culture. In this evaluation, symbolic reduction is avoided by grounding the discussion at the tectonic level, that is, in the physical characteristics of intervals between columns, fittings, and engawa. This belongs to theories of Japanese architecture. (1978), (1999). The 1978 exhibition was contemporaneous with structuralist theories of Japan such as Roland Barthes, *L'Empire des signes* (1970). In international presentations of Japanese architecture, points of contact with Norberg-Schulz's theory of place have also been discussed.

5.7 Computational morphogenesis: form generation through materials and computation (Menges)

The theory "Computational morphogenesis: form generation through materials and computation (Menges)," associated with Menges, A. at the Institute for Computational Design and Construction (ICD), Stuttgart , and Oxman, N. at the MIT Media Lab Mediated Matter Group, advances the idea of material computation: the inherent properties of materials, such as fiber orientation, elasticity, and moisture content, are integrated into computational models, and the "behavior" of materials is treated as an active participant in the design process (Menges, 2012, 2015). Form is not decided arbitrarily by the designer, but emerges from the interaction among material properties, fabrication constraints, structural mechanics, and environmental response.

Material computation extends Frei Otto's form-finding into computational environments and belongs to a lineage that incorporates the inherent properties of materials as active participants. The structure Field (the latent property space of the material) → Wave (material response to environmental change) → Edge (the interface between material and fabrication) → Vortex (the

emergence of form) → Bundle (the institutionalization of design-fabrication processes) is the computational version of Otto's correspondence with the five stages. It also shows structural similarity to reaction-diffusion patterns in D03 chemistry: self-shaping through changes in material moisture content is an architectural-scale version of chemical self-organization. Because it draws upon principles of biological morphogenesis, it also connects to D06 evolutionary biology and D10 developmental biology.

What is distinctive in Menges is the formulation that materials act. It is neither that the designer decides form nor that physical law alone determines form, but that the inherent properties of materials actively participate in morphogenesis. This is computational-architectural evidence that Field, as the latency of material, is not merely a passive substrate but an active participant in generation. This belongs to computational design and digital fabrication. Menges / ICD Stuttgart (2008-), Oxman / MIT (2010-). The lineage from Frei Otto's IL is explicitly inherited. The Research Pavilions series (2010-present) is the representative body of work. It also connects with research centers such as the AA Design Research Lab and ETH Zurich.

5.8 Half-house / incremental architecture: progressive generation over time (Aravena / ELEMENTAL)

The theory "Half-house / incremental architecture: progressive generation over time (Aravena / ELEMENTAL)," associated with Aravena, A. / ELEMENTAL. Quinta Monroy (2004, Iquique, Chile), proposes the half-house (half a good house): a strategy in which public funding supplies one high-quality half of a dwelling, while residents expand the remaining half through their own efforts (Aravena & Iacobelli, 2012, *ELEMENTAL: Incremental Housing and Participatory Design Manual*). This "half" is not an arbitrary half. Rather, professionals provide those parts that residents cannot easily produce on their own, such as the structural frame, service core, and shared areas, while leaving to residents those parts they can build themselves, such as partitions, finishes, and extension space.

The half-house is an implementation of the five stages on the time axis. Professionals provide the Field, that is, the framework, and residents carry out the process of Wave to Edge to Vortex to Bundle over time. This structure is a social implementation of Alexander's unfolding, or stepwise generation, and it also gives architectural embodiment to the continuous transformation of the five stages as a design that does not finish. The interface between the first "half" and the second "half" is a temporal design of Edge. The joint between the structural frame and the extension space is a physical Edge at which residents' acts of living generate form over time.

What is distinctive in Aravena is the design of incompleteness. While other architectural theories aim at a finished form, Aravena positively adopts "not being complete" as a design strategy. This corresponds to the fact that the five stages describe a return from Bundle to Field, that is, a circulation, and it is an architectural expression of a process in which no final completion exists. This belongs to social architecture. Aravena / ELEMENTAL (2001-). He received the Pritzker Prize in 2016. Beginning from the context of Chilean social housing policy, this work developed internationally as a methodology of incremental design. Including the direction of the 2016 Venice Biennale ("Reporting from the Front"), it forms part of a movement to redefine the social role of architecture.

5.9 Bodily sensation and architecture: phenomenology of tactile perception and spatial experience (Pallasmaa)

The theory "Bodily sensation and architecture: phenomenology of tactile perception and spatial experience (Pallasmaa)," presented in Pallasmaa, J. (1996/2005) *The Eyes of the Skin: Architecture and the Senses*. Wiley, argues that modern architecture has been biased toward the primacy of vision, that is, ocularcentrism, and has neglected the perception of space through the

whole body, including touch, hearing, and smell (Pallasmaa, 1996, 2005 2nd ed.). Architectural experience is constituted not by vision alone but as multisensory integration across the entire body. Touch precedes vision as "the most primordial sense."

Pallasmaa's concept of atmosphere can be read as a perceptual account of Stage 1 in the five-stage model, namely Field understood as drift or undivided ambience. The overall quality of space as constituted through multisensory integration is an architectural version of a state in which things have not yet separated, like an undifferentiated sea. Immersion in space through peripheral vision is worth attention as a possible perceptual mechanism for the experience of being in Field. The shift toward focused vision, that is, objectification, may correspond to Wave as the emergence of difference.

What is distinctive in Pallasmaa is the distinction between modes of perception, especially peripheral and central vision. This connects to the question in D2 of how lack is detected under different perceptual modes. It suggests the possibility of a process in which the detection of predictive error begins peripherally, holistically, bodily, and preconsciously, and is then clarified centrally, analytically, and consciously. This is promising as an exploratory hypothesis, but it exceeds the scope of the present evaluation. This belongs to architectural phenomenology. Pallasmaa (1996, 2005 2nd ed.). Its lineage comes from the phenomenology of embodiment in Merleau-Ponty. It shares an intellectual context with Norberg-Schulz's theory of place and Heidegger's ontology. It is widely read in architectural education. There is also a parallel with Zumthor (*Atmospheres*, 2006).

5.10 Parametricism: continuous differentiation and the logic of fields (Schumacher)

The theory "Parametricism: continuous differentiation and the logic of fields (Schumacher)," presented in Schumacher, P. (2008) "Parametricism as Style — Parametricist Manifesto" (presented at Venice Biennale 2008); Schumacher, P. (2011-2012) *The Autopoiesis of Architecture*, 2 vols., Wiley, proposed parametricism as a new architectural style after postmodernism. It links all elements of design parametrically and generates continuous differentiation (Schumacher, 2008). Its heuristics are as follows: (1) avoid rigid geometric primitives, (2) avoid repetition, and (3) avoid unmediated juxtaposition, so that everything changes continuously and takes on gradient.

The "continuous differentiation" of parametricism has a superficial similarity to Stage 2 of the five-stage model, Wave as fluctuation of difference, insofar as it can be read as its computational amplification. The claim that "all parameters are continuously related" also creates a point of contact with the concept of field, that is, Stage 1 as Field. However, a strict critical standard must be applied here. Continuous differentiation generates difference in a smooth manner, and this differs from the quality of opposition and separation described by Wave in the five-stage model. In the five-stage model, Wave means that differences arise within Field and appear as oscillation, and this includes tension and opposition. The continuous differentiation of parametricism moves instead toward dissolving tension through smooth continuity, and is therefore oriented in the opposite direction from the structure described by Wave in the five-stage model.

The final judgment is that parametricism contains a decisive structural contradiction with Edge, a central concept of the five-stage model: it tends toward erasing boundaries rather than toward events that arise at boundaries. Partial correspondences with Field and Wave are not enough to justify a judgment of structural similarity. This belongs to architectural style theory. Schumacher (2008, 2011-2012). As design director of Zaha Hadid Architects, he has practical influence. As architectural theory, the validity of its use of Luhmann's theory has been widely criticized. Its politics as a "style manifesto" has also been a matter of debate.

5.11 poché: wall thickness as the material band of boundary

The theory "poché: wall thickness as the material band of boundary," a representational technique and concept in architecture since the Beaux-Arts tradition, developed in urban theory by Rowe, C. & Koetter, F. (1978) *Collage City*, MIT Press, treats poché as a term indicating thickness, wall, and residue. It extends into urban theory as urban poché (Rowe & Koetter, 1978). Boundary is not a "line" but a "thick space," and internalized residue, such as service zones, semi-outdoor areas, and the depth of openings, constitutes the experience of city and architecture.

Poché is the material implementation of Edge. "Wall thickness as the band of boundary" is an architectural confirmation of the anthropological proposition in D25 AN-001 that transitional phases are bands. Internalized residue, such as storage, equipment, and openings, is the material version of the internal structure of Edge. This corresponds directly to Boundaries in Alexander, that is, thickened outlines, but whereas Alexander locates Boundaries within a system as one of the fifteen properties of centers, poché shows a more concrete implementation as design operation and representational technique.

What appears here is the materiality of Edge. While other theories speak of Edge conceptually or socially, poché treats Edge as physical thickness. Its expansion into urban poché demonstrates the trans-scalar movement of Edge: wall thickness at the level of detail, façade at the scale of the building, and urban mass at the scale of the city. This belongs to architectural design and architectural drawing. It derives from the Beaux-Arts tradition. Its development into urban theory is visible in Rowe & Koetter (1978) *Collage City*. In recent years, practices that positively design wall thickness as inhabitable place have increased under the label of thick wall architecture.

5.12 edge: urban image boundaries and seams (Lynch)

The theory "edge: urban image boundaries and seams (Lynch)," presented in Lynch, K. (1960) *The Image of the City*, MIT Press, argues that the image of the city can be described through five elements: paths, edges, districts, nodes, and landmarks, and that edges orient memory and action as boundaries and seams (Lynch, 1960). An edge is a linear element that is not a path and can become either a barrier or a seam. This ambivalence is the core of Lynch's concept of edge.

The two-sidedness of edge, which can be either barrier or seam, is the urban-scale version of the ambivalence of Edge. Lynch confirms at the level of the city the pair found in D25 anthropology between Douglas, where pollution is a destructive boundary, and Turner, where liminality is a generative boundary. Jacobs's border vacuum complements Lynch's edge as an urban-theoretical description of the city in a state where edge has died.

What is distinctive here is the methodology by which Edge is extracted from citizens' experience, namely cognitive mapping. Rather than relying only on theoretical concepts, Lynch offers a framework for describing the Edge actually experienced by urban residents. Whereas other D27 entries describe Edge from the standpoint of designers and theorists, Lynch describes Edge through users' cognition. This belongs to urban planning and environmental psychology. Lynch (1960). It is foundational for research on cognitive maps, and has generated many quantitative extensions. It also had methodological influence on Space Syntax (Hillier & Hanson, 1984). Research on urban legibility continues to develop through integration with GIS and machine learning.

6. Cross-Cutting Patterns

First, as shown by Pattern Language / Nature of Order: mutual support among centers and unfolding (Alexander), Finding Form: exploration of equilibrium forms through physical models (Frei Otto), and in-between realm / doorstep: boundary as mediation between binary oppositions (van Eyck), this domain describes difference and opposition not merely as obstacles but as

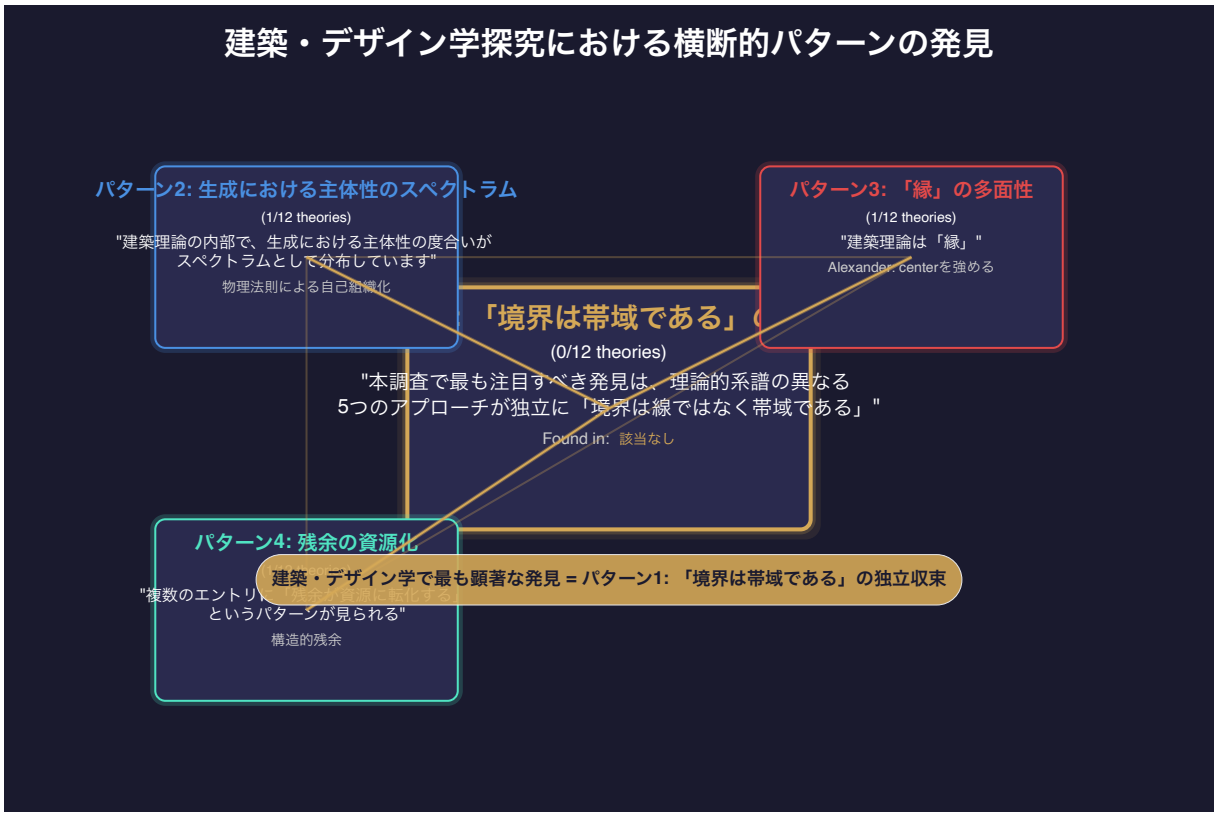

Figure 3: Boundary Design in Architecture and Spatial Design and Its Correspondence to the Five-Stage Model —

conditions for the rise of new order. This means that, within the five stages, the transition from Edge to Vortex is especially richly observable in this domain.

Second, what competence / threshold space: designing gradients between public and private (Hertzberger), Conditions of urban diversity: boundary permeability and the ecology of streets (Jacobs), and Spatial theory of (ma): an architectural account of Japanese space-time share is that generation is not a one-off event. It includes a circulation involving the formation of prior conditions, the amplification of tension, local ordering, stabilization, and subsequent reconfiguration. This confirms that the five-stage model must be read not as a static table of classification but as a temporal movement.

At the same time, Bodily sensation and architecture: phenomenology of tactile perception and spatial experience (Pallasmaa) and Parametricism: continuous differentiation and the logic of fields (Schumacher) also show that correspondence with the five stages does not arise automatically. A match in the number of stages or in the similarity of terms is not sufficient. It is necessary to examine case by case which concepts carry the transition point and at what level structure comes into being.

7. Unresolved Questions

1. To what extent can Bodily sensation and architecture: phenomenology of tactile perception and spatial experience (Pallasmaa) and Parametricism: continuous differentiation and the logic of fields (Schumacher) be read as stage-descriptions of the five-stage model?

2. To what extent can the structure confirmed in this domain be extended from the explanation of individual cases into a more general model of generation? Further scrutiny is still needed regarding the conditions under which comparability can be maintained without losing the

specificity of the object.

3. Should the five stages be read as a linear progression, or as a circulation involving multiple temporal layers and repetitions? In particular, theories differ in how explicitly they indicate the return from Bundle to the next Field.

4. What becomes visible, and what is lost, when the domain-specific vocabularies of this field are translated into the general vocabulary of the five stages? The boundary between translatability and untranslatability is itself an important issue for future comparative inquiry.

8. Conclusion

This renewed examination of the domain confirmed strong structural correspondence in ten of the twelve theories and conditional correspondence in one. In particular, Pattern Language / Nature of Order: mutual support among centers and unfolding (Alexander), Finding Form: exploration of equilibrium forms through physical models (Frei Otto), and in-between realm / doorstep: boundary as mediation between binary oppositions (van Eyck) provide especially detailed descriptions of either the movement from Field to Edge or the movement from Edge to Vortex within the five stages, and can therefore be positioned as core evidence within this domain.

At the same time, the findings of this domain do not claim that the same five stages appear unchanged in every theory. The stages that stand out most strongly differ from theory to theory, and some cases require reservation in stage mapping. What matters, therefore, is not to generalize directly from the high correspondence rate, but to continue comparing which aspects are described in which vocabularies.

In that sense, this domain not only supports the validity of the five-stage model in one direction, but also sets constraints on how the model should be read. The very fact that this domain contains both theories with dense correspondence and theories requiring reservation constitutes the value of the comparative investigation.